

Q1 For the following sequence of pairs find the hit ratio using (i) FIFO (ii) LRU (iii) MRU (iv) optimal page replacement algorithm with physical memory of 3 frames

FIFO:

3	2	1	3	4	1	6	2	4	3	4	2	1	4	5	2	1	3	4
3	3	3	3	4	4	6	6	4	3	4	4	4	4	5	5	5	3	4
-	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
-	-	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
x	x	x	v	x	v	x	v	x	x	x	x	v	v	x	v	v	x	x

Hits = 8 faults = 11

Hit ratio = $8/19 = 42.11\%$

(ii) MRU: LRU:

3	2	1	3	4	1	6	2	4	3	4	2	1	4	5	2	1	3	4
3	3	3	3	3	3	6	6	6	3	3	3	1	1	1	2	2	2	4
-	2	2	2	4	4	4	2	2	2	2	2	2	2	5	5	5	3	3
-	-	1	1	1	1	1	1	4	4	4	4	4	4	4	4	1	1	1
x	x	x	v	x	v	x	x	x	x	v	v	x	x	x	x	x	x	x

Hits = 5 faults = 14

Hit ratio = $5/19 = 26.32\%$

(iii) MRU:

3	2	1	3	4	1	6	2	4	3	4	2	1	4	5	2	1	3	4
3	3	3	3	4	4	4	4	4	3	4	4	1	4	5	2	2	2	2
-	2	2	2	2	2	2	2	2	2	2	2	4	1	1	1	1	3	4
-	-	1	1	1	1	6	6	6	6	6	6	6	6	6	6	6	6	6
x	x	x	v	x	v	x	v	v	x	x	v	x	v	x	x	v	x	x

H = 7 F = 12

$\Rightarrow 7/19 = 36.84\%$

(iv) optimal:

3	2	1	3	4	1	6	2	4	3	4	2	1	4	5	2	1	3	4
3	3	3	3	4	4	4	4	4	4	4	4	4	4	5	5	5	3	4
-	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
-	-	1	1	1	1	6	6	6	3	3	3	1	1	1	1	1	1	1
F	x	x	✓	x	✓	x	✓	✓	x	✓	✓	✓	✓	x	✓	✓	x	x

Hits = 9 Faults = 10

Hit ratio = $9/19 = 47.37\%$ Efficiency:

Optimal Page Replacement is the most efficient - for this reference string because it produces the fewest page faults (10) and highest hit ratio (47.37%).

Belady's Anomaly:

Algorithm	Belady's Anomaly
FIFO	Yes
LRU	NO
Optimal	NO
MRU	depends on reference behaviour.

Apply FIFO page replacement algo and check whether belady's anomaly exists or not. Physical memory is 3 and 4 frames.

1	2	3	4	1	2	5	1	2	3	4	5
1	1	1	4	4	4	5	5	5	3	3	3
-	2	2	2	1	1	1	1	1	1	4	4
-	x	3	3	3	2	2	2	2	2	2	5
x	x	x	x	x	x	x	x	✓	✓	x	✓

Page faults = 9

$$\text{Hit} = \frac{3}{12} \times 100 = 25\%$$

1	2	3	4	1	2	5	1	2	3	4	5
1	1	1	1	1	1	5	5	5	5	1	1
-	2	2	2	2	2	2	1	1	1	1	2
-	-	3	3	3	3	3	3	3	3	3	3
-	-	-	4	4	4	4	4	2	2	4	4
x	x	x	x	✓	✓	x	x	x	x	x	x

faults = 10

hits = 2

$$\text{Hit ratio} = \frac{2}{12} \times 100 = 16.67\%$$

Belady's Anomaly: Since increasing the physical memory from 3 frames to 4 frames increases page faults from 9 to 10, Belady's Anomaly exists.

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Efficiency:

FIFO takes $O(n \times t)$ time and $O(t)$ space, where n = no. of pages and t = no. of frames.

It checks the pages in the frames to find a hit and replaces the oldest page first using a queue.